Learning objectives

1. To learn about the types, importance of the electrical system used in an automobile.
2. To understand the construction, working principle of various electrical systems used in an automobile.

10.0 Introduction

Electrical energy is a form of an energy. Electrical energy is used for domestic, agriculture and industrial purpose etc. Electrical energy plays an important role in an automobile. The automotive electrical system can be classified as follows.

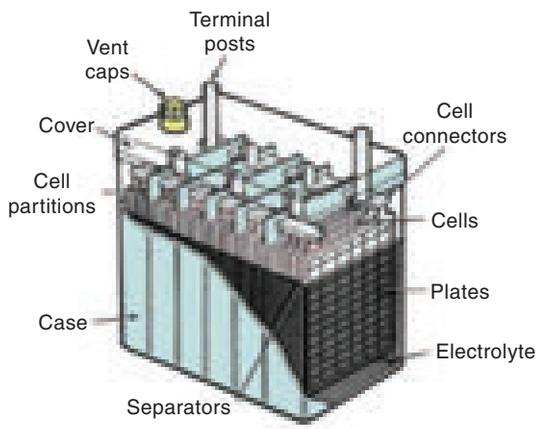
1. Starting System
2. Ignition System
3. Lighting System
4. Generation, Storage and Distribution System
5. Accessories

10.1 Battery

The battery is an important and essential component in an automotive electrical system. The electrical energy is stored in the battery and it is supplied to all the electrical subsystem when the vehicle is not operating. The battery is considered as the heart of an automotive electrical system.

A positive terminal and a negative terminal form a cell. Several cells are connected in series or in parallel is called as a battery.

A battery is used to store electrical energy as chemical energy. This chemical energy is then converted to electrical energy as and when required. The conversion of electrical energy into chemical energy by applying external electrical source is known as charging of the battery. Whereas conversion of chemical energy into electrical energy for supplying the external load is known as discharging of the battery.



10.1.1 Types of battery

1. Lead Acid Battery
2. Lithium-iron Battery
3. Nickel Cadmium Battery
4. Nickel Metal Hydride Battery
5. Sodium Sulphur Battery

Out of which, the lead-acid battery is widely used in an automobile.

10.1.2 Lead Acid Battery Construction And Working Principle

The following are the important parts of a battery

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